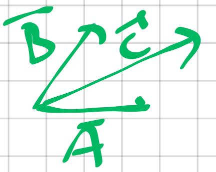
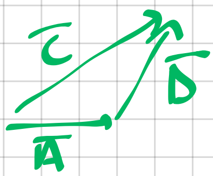
funda-

→ unit vector: $\hat{a} = \frac{\bar{A}}{|\bar{A}|}$

→ $\bar{A}+\bar{B}=\bar{C}$

(1) parallelogram rule (2) head to tail rule





→ $\bar{A}+\bar{B}=\bar{B}+\bar{A}$ commutative

$\bar{A}+(\bar{B}+\bar{C})=(\bar{A}+\bar{B})+\bar{C}$ associative

→ dot prod. $\bar{A}\cdot\bar{B}=AB\cos\theta_{AB}$

cross prod. $\bar{A}\times\bar{B}=AB\sin\theta_{AB}$

→ scalar triple prod.

$\bar{A}\cdot(\bar{B}\times\bar{C})=\bar{B}\cdot(\bar{C}\times\bar{A})=\bar{C}\cdot(\bar{A}\times\bar{B})$

→ vector triple prod.

$\bar{A}\times(\bar{B}\times\bar{C})=\bar{B}\times(\bar{C}\times\bar{A})=\bar{C}\times(\bar{A}\times\bar{B})$

→ unit vectors.

$\hat{a_x},\hat{a_y},\hat{a_z} \xrightarrow{\text{in}} (x,y,z)$

$\hat{a_x}\cdot\hat{a_x}=\hat{a_y}\cdot\hat{a_y}=\hat{a_z}\cdot\hat{a_z}=1$

$\hat{a_x}\cdot\hat{a_y}=\hat{a_y}\cdot\hat{a_z}=\hat{a_x}\cdot\hat{a_z}=0$

$\hat{a_x}\times\hat{a_y}=\hat{a_z}$

$\hat{a_y}\times\hat{a_z}=\hat{a_x}$

$\hat{a_z}\times\hat{a_x}=\hat{a_y}$

right hand thumb rule

(1) Cartesian coord. sys.

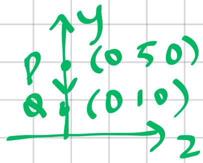
→ range: $-\infty<(x,y,z)<\infty$

→ $\bar{A}=A_x\hat{a_x}+A_y\hat{a_y}+A_z\hat{a_z}$

→ diff. length

$\bar{dl}=dx\,\hat{a_x}+dy\,\hat{a_y}+dz\,\hat{a_z}$

eg:



$\bar{dl}=dy\,\hat{a_y}$

$\int\bar{dl}=\left[\int_5^1 dy\,\hat{a_y}\right]$

$\bar{PQ}=y|_5^1\,\hat{a_y}$

$\bar{PQ}=-4\hat{a_y}$

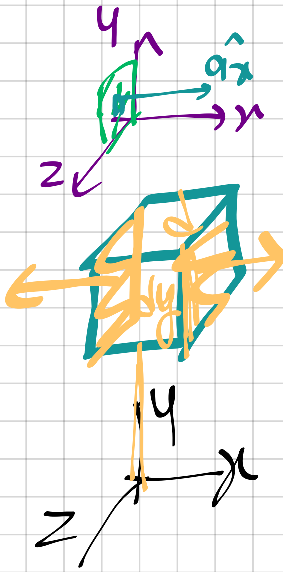
#OR $\int_1^5 dy\,(-\hat{a_y})$

→ diff. surface

$\bar{ds_x}=dy\,dz\,\hat{a_x}$

$\bar{ds_y}=dx\,dz\,\hat{a_y}$

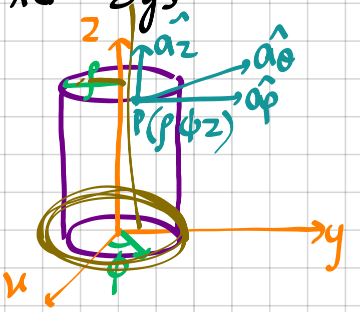
$\bar{ds_z}=dx\,dy\,\hat{a_z}$



→ diff. volume

$dV=dx\,dy\,dz$

(2) cylindrical coord sys:

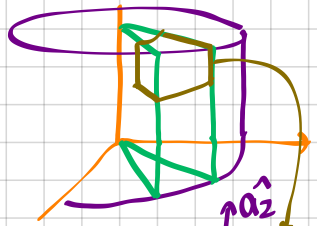


→ range:

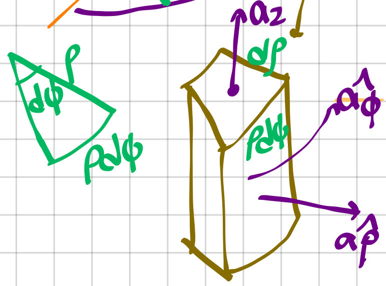
$0<\rho<\infty$
$0<\phi<2\pi$
$-\infty<z<\infty$

→ diff. length

$\vec{dl} = d\rho\,\hat{a}_\rho + \rho d\phi\,\hat{a}_\phi + dz\,\hat{a}_z$



→ diff. surface

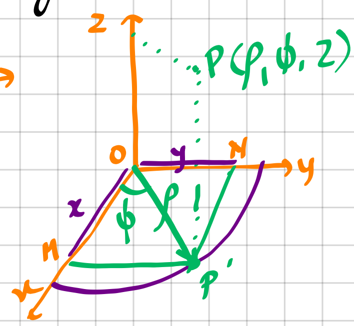


$\vec{dS}_\rho = \rho d\phi\, dz\,\hat{a}_\rho$
$\vec{dS}_\phi = d\rho\, dz\,\hat{a}_\phi$
$\vec{dS}_z = \rho d\rho\, d\phi\,\hat{a}_z$

→ diff. volume: $dV = \rho d\rho d\phi dz$

* cylindrical ⇄ cartesian

→



$$\left\{\begin{array}{l} x=\rho\cos\phi \;① \\ y=\rho\sin\phi \;② \\ z=z \end{array}\right. \quad \left.\begin{array}{ll} \rho=\sqrt{x^2+y^2} & ①^2+②^2 \\ \phi=\tan^{-1}\left(\frac{y}{x}\right) & \frac{②}{①} \end{array}\right\}$$

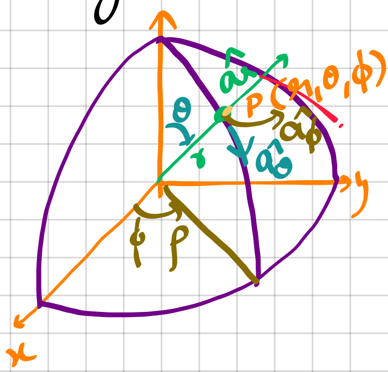
→ transformation of vectors

d/dφ   d/dz → chumma to rem ashte

$$\begin{bmatrix}A_x\\A_y\\A_z\end{bmatrix}=\begin{bmatrix}\cos\phi & -\sin\phi & 0\\ \sin\phi & \cos\phi & 0\\ 0 & 0 & 1\end{bmatrix}\begin{bmatrix}A_\rho\\A_\phi\\A_z\end{bmatrix}$$

$$\begin{bmatrix}A_\rho\\A_\phi\\A_z\end{bmatrix}=\begin{bmatrix}\cos\phi & \sin\phi & 0\\ -\sin\phi & \cos\phi & 0\\ 0 & 0 & 1\end{bmatrix}\begin{bmatrix}A_x\\A_y\\A_z\end{bmatrix}$$

(3) spherical coord. sys.

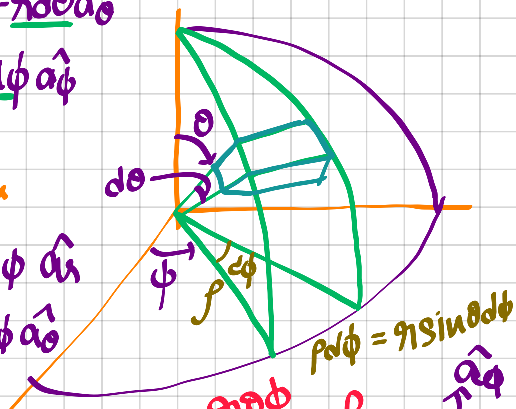


→ range:

$0<r<\infty$
$0<\theta<\pi$
$0<\phi<2\pi$

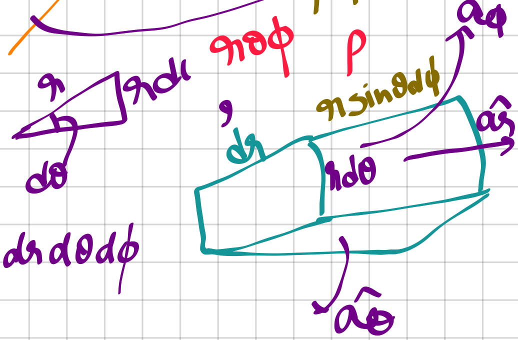
→ diff. length:

$\vec{dl} = dr\,\hat{a}_r + r d\theta\,\hat{a}_\theta + r\sin\theta d\phi\,\hat{a}_\phi$



→ diff. surface

$\vec{dS}_r = r^2\sin\theta d\theta d\phi\,\hat{a}_r$
$\vec{dS}_\theta = r\sin\theta dr\, d\phi\,\hat{a}_\theta$
$\vec{dS}_\phi = r dr d\theta\,\hat{a}_\phi$

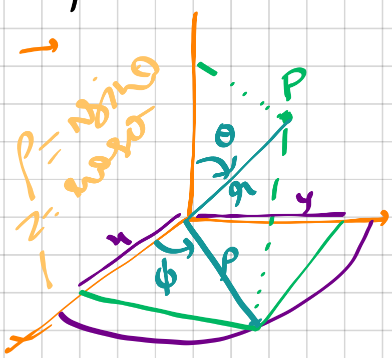


→ diff. volume

$dV = r^2\sin\theta\, dr\, d\theta\, d\phi$

→ spherical ⇄ cartesian

→



$$\left\{\begin{array}{l} x=\rho\cos\phi \\ \quad = r\sin\theta\cos\phi \;① \\ y=\rho\sin\phi \\ \quad = r\sin\theta\sin\phi \;② \\ z=r\cos\theta \;③ \end{array}\right.$$

$$\left.\begin{array}{ll} r=\sqrt{x^2+y^2+z^2} & ①^2+②^2+③^2 \\ \phi=\tan^{-1}\left(\frac{y}{x}\right) & \frac{②}{①} \\ \theta=\cos^{-1}\left(\frac{z}{r}\right) & \text{frm } 3 \end{array}\right\}$$

→ transformation of vectors

d/dθ   d/dφ → chumma

$$\begin{bmatrix}A_x\\A_y\\A_z\end{bmatrix}=\begin{bmatrix}\sin\theta\cos\phi & \cos\theta\cos\phi & -\sin\phi\\ \sin\theta\sin\phi & \cos\theta\sin\phi & \cos\phi\\ \cos\theta & -\sin\theta & 0\end{bmatrix}\begin{bmatrix}A_r\\A_\theta\\A_\phi\end{bmatrix}$$

$$\begin{bmatrix}A_r\\A_\theta\\A_\phi\end{bmatrix}=\begin{bmatrix}\sin\theta\cos\phi & \sin\theta\sin\phi & \cos\theta\\ \cos\theta\cos\phi & \cos\theta\sin\theta & -\sin\theta\\ -\sin\phi & \cos\phi & 0\end{bmatrix}\begin{bmatrix}A_x\\A_y\\A_z\end{bmatrix}$$

✱ Del operator (∇)

(1) Gradient [scalar → vector]

$$\nabla_{ct} V = \frac{\partial V}{\partial x}\hat{a}_x + \frac{\partial V}{\partial y}\hat{a}_y + \frac{\partial V}{\partial z}\hat{a}_z$$

$$\nabla_{cy} V = \frac{\partial V}{\partial \rho}\hat{a}_\rho + \frac{1}{\rho}\frac{\partial V}{\partial \phi}\hat{a}_\phi + \frac{\partial V}{\partial z}\hat{a}_z$$

$$\nabla_{sp} V = \frac{\partial V}{\partial r}\hat{a}_r + \frac{1}{r}\frac{\partial V}{\partial \theta}\hat{a}_\theta + \frac{1}{r\sin\theta}\frac{\partial V}{\partial \phi}\hat{a}_\phi$$

(2) Divergence [vector → scalar]

ct: $$\nabla\cdot\bar{V} = \frac{\partial V_x}{\partial x} + \frac{\partial V_y}{\partial y} + \frac{\partial V_z}{\partial z}$$

cy: $$\nabla\cdot\bar{V} = \frac{1}{\rho}\frac{\partial(\rho V_\rho)}{\partial \rho} + \frac{1}{\rho}\frac{\partial V_\phi}{\partial \phi} + \frac{\partial V_z}{\partial z}$$

sp: $$\nabla\cdot\bar{V} = \frac{1}{r^2}\frac{\partial(r^2 V_r)}{\partial r} + \frac{1}{r\sin\theta}\frac{\partial(\sin\theta\, V_\theta)}{\partial \theta} + \frac{1}{r\sin\theta}\frac{\partial V_\phi}{\partial \phi}$$

(3) Divergence theorem:

$$\oint_S A\cdot ds = \int_V (\nabla\cdot\bar{A})\, dV$$

(4) Curl of vector [vector → vector]

ct: $$\nabla\times\bar{V} = \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ V_x & V_y & V_z \end{vmatrix}$$

cy: $$\nabla\times\bar{V} = \frac{1}{\rho}\begin{vmatrix} \hat{a}_\rho & \rho\hat{a}_\phi & \hat{a}_z \\ \partial/\partial \rho & \partial/\partial \phi & \partial/\partial z \\ V_\rho & \rho V_\phi & V_z \end{vmatrix}$$

sp: $$\nabla\times\bar{V} = \frac{1}{r^2\sin\theta}\begin{vmatrix} \hat{a}_r & r\hat{a}_\theta & r\sin\theta\,\hat{a}_\phi \\ \partial/\partial r & \partial/\partial \theta & \partial/\partial \phi \\ V_r & rV_\theta & r\sin\theta\, V_\phi \end{vmatrix}$$

(5) Stokes theorem

$$\oint_l \bar{A}\cdot d\bar{l} = \int_S (\nabla\times\bar{A})\cdot d\bar{S}$$

$$d\bar{l} \xrightarrow{\nabla\times A} d\bar{S} \xrightarrow{\nabla\cdot A} d\bar{V}$$

stoke   div

Imp. pts.

(1) Solenoid $\nabla\cdot\bar{V}=0$

(2) irrotational / conservative $\nabla\times\bar{V}=0$

(3) harmonic $\nabla^2 V = 0$

(4) curl(grad v) = 0 CG=0

(5) div(curl v) = 0 DC=0

(6) Laplacian

ct: $$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}$$

cy: $$\nabla^2 V = \frac{1}{\rho}\frac{\partial}{\partial \rho}\left(\rho\frac{\partial V}{\partial \rho}\right) + \frac{1}{\rho}\frac{\partial^2 V}{\partial \phi^2} + \frac{\partial^2 V}{\partial z^2}.$$

$$\nabla^2 V = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial V}{\partial r}\right) + \frac{1}{r^2\sin\theta}\left(\frac{\partial}{\partial\theta}\sin\theta\frac{\partial V}{\partial\theta}\right) + \frac{1}{r^2\sin\theta}\frac{\partial^2 V}{\partial\phi^2}$$

Electrostatics

→ Coulombs law.

$$F = \frac{K Q_1 Q_2}{R^2} \qquad \left[K = \frac{1}{4\pi\varepsilon_0}\right]$$

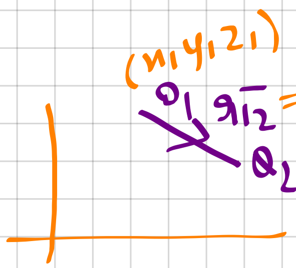
$$\varepsilon_0 = 8.854 \times 10^{-12}\ F/m$$

$$\varepsilon_0 = \frac{10^{-9}}{36\pi}\ F/m$$

$$K = 9 \times 10^9$$

$$\bar{F}_{12} = \frac{K Q_1 Q_2}{R_{12}^2}\hat{a}_{R_{12}}$$

$$\boxed{\bar{F}_{12} = \frac{K Q_1 Q_2}{R_{12}^3}\bar{R}_{12}}$$



$$\bar{R}_{12} = (x_2 - x_1)\hat{a}_x + (y_2 - y_1)\hat{a}_y + (z_2 - z_1)\hat{a}_z$$

→ $\bar{F}_{12} = -\bar{F}_{21}$

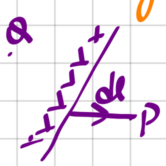
→ Electric field intensity

$$\bar{E} = \frac{\bar{F}}{q}$$

→ $\bar{F} = \frac{K Q q (\bar{r} - \bar{r}_1)}{(\bar{r} - \bar{r}_1)^3}$

→ $\bar{E}_{net} = \frac{K Q_1 (\bar{R} - \bar{R}_1)}{|\bar{R} - \bar{R}_1|^3} + \frac{K Q_2 (\bar{R} - \bar{R}_2)}{|\bar{R} - \bar{R}_2|^3}$

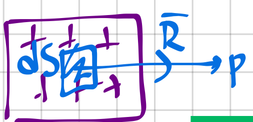
→ Charge dist.



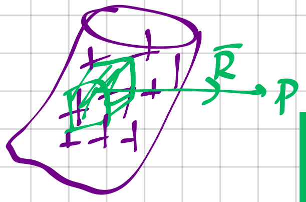
$$dQ = \rho_L dl$$

$$Q = \int \rho_L dl$$

$$\boxed{E = \frac{KQ}{R^2} = k\frac{\int_L \rho_L dl}{R^2}\hat{a}_R}$$



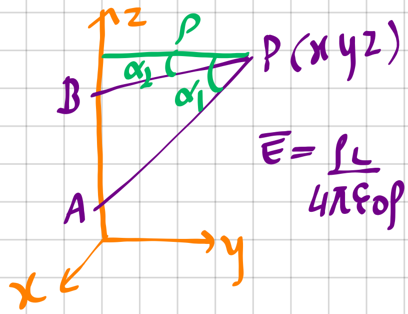
$dQ = \rho_s dS$; $\boxed{\bar{E} = k\frac{\int_S \rho_s dS}{R^2}\hat{a}_R}$



$$dQ = \rho_v dV$$

$$\boxed{E = K\frac{\int_V \rho_v dV}{R^2}\hat{a}_R}$$

→ $\bar{E}$ due to finite length line charge

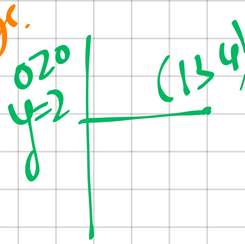


$$\bar{E} = \frac{\rho_L}{4\pi\varepsilon_0\rho}\left[(\cos\alpha_2 - \cos\alpha_1)\hat{a}_z - (\sin\alpha_2 - \sin\alpha_1)\hat{a}_\rho\right]$$

C1: infinite long line charge.

$\alpha_1 = 90°,\ \alpha_2 = -90°$

$$\boxed{E = \frac{\rho_L}{2\pi\varepsilon_0\rho}\hat{a}_\rho} = \frac{\rho_L}{2\pi\varepsilon_0}\frac{\bar{\rho}}{\rho^2}$$



C2: when pt of intersection is bisector of line charge.

$\alpha_2 = -\alpha_1$

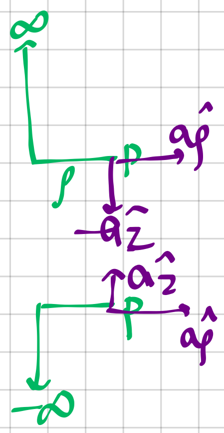
$$\bar{E} = \frac{\rho_L}{2\pi\varepsilon_0\rho}\sin\alpha_1\,\hat{a}_\rho$$

C3: when pt of intersect is in one end of infinite line charge.

$\alpha_1 = 0,\ \alpha_2 = -90°$

$$E = \frac{\rho_L}{4\pi\varepsilon_0\rho}(\hat{a}_\rho - \hat{a}_z)$$

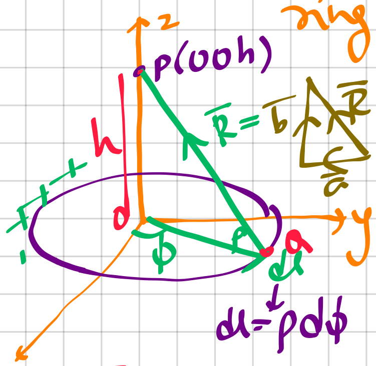
$$E = \frac{\rho_L}{4\pi\varepsilon_0\rho}(\hat{a}_\rho + \hat{a}_z)$$



→ Electric field due to circular ring

$$\bar{E} = k \frac{\int \rho_L \, dl}{R^3} \bar{R}$$

$$= k \frac{\int \rho_L (\rho \, d\phi) \, \bar{R}}{|R|^3}$$



by symm a_ρ comp cancel out

$\bar{QP} = \bar{QO} + \bar{OP}$

$\bar{R} = -\rho \hat{a}_\rho + h \hat{a}_z$

$$= k \frac{\int_0^{2\pi} \rho_L (\rho \, d\phi) [\cancel{-\rho \hat{a}_\rho} + h \hat{a}_z]}{(\rho^2 + h^2)^{3/2}}$$

$$= \frac{1}{4\pi\varepsilon_0} \frac{\rho_L \rho \int_0^{2\pi} h \hat{a}_z}{(\rho^2+h^2)^{3/2}} \qquad \int_0^{2\pi} d\phi = 2\pi$$

$$= \frac{\rho_L \rho h \, \hat{a}_z (2\pi)}{4\pi\varepsilon_0 (\rho^2+h^2)^{3/2}}$$

$$\boxed{\bar{E} = \frac{\rho_L \rho h}{2\varepsilon_0 (\rho^2+h^2)^{3/2}} \hat{a}_n}$$

→ E due to circular disk.

$$\bar{E} = \frac{\rho_s}{2\varepsilon_0} \left[1 - \frac{h}{\sqrt{R^2+h^2}}\right] \hat{a}_z$$

if infinitely long disk ≡ infinite sheet

$$\boxed{E = \frac{\rho_s}{2\varepsilon_0} \hat{a}_n}$$

→ Electric flux.

$$\boxed{\psi = \int_S \bar{D} \cdot \bar{dS}} \qquad \boxed{\bar{D} = \varepsilon \bar{E}}$$

→ Gauss law

$$\psi = \oint_S \bar{D} \cdot \bar{dS} = Q_{enc} = \int_V \rho_v \, dv$$

$$\boxed{\oint_S \bar{D} \cdot \bar{dS} = \int_V \rho_v \, dv}$$ Maxwells 1st eqn (Integral form)

→ $\oint \bar{D} \cdot \bar{dS} = \int_V (\nabla \cdot \bar{D}) \, dV$ (from divergence theorem)

$$\oint \bar{D} \cdot \bar{dS} = \int_V \rho_v \, dV$$

$$\int_V (\nabla \cdot \bar{D}) \, dV = \int_V \rho_v \, dV$$

$$\boxed{\nabla \cdot \bar{D} = \rho_v}$$ Maxwells 1st eqn (differential/point form)

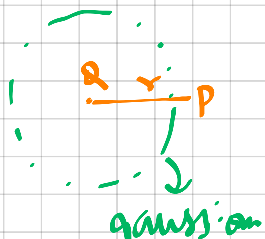
→ $\bar{E}$ due to pt charge

$$\oint \bar{D} \cdot \bar{dS} = Q_{enc}$$

$$\varepsilon \oint \bar{E} \cdot \bar{dS} = Q$$

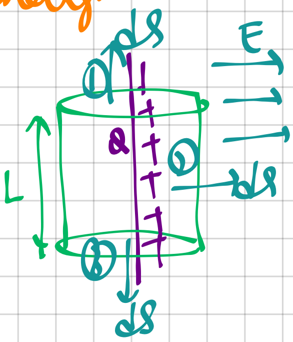
$$\varepsilon E (4\pi r^2) = Q$$

$$\boxed{E = \frac{Q}{4\pi\varepsilon r^2}}$$



$\oint dS = 4\pi r^2$

→ $\bar{E}$ due to line charge



$$\oint \bar{D} \cdot \bar{dS} = Q_{enc}$$

$$\varepsilon \oint \bar{E} \cdot \bar{dS} = Q$$

$$\varepsilon \left[\cancel{\oint_1 \bar{E} \cdot \bar{dS}} + \oint_2 + \oint_3 \right] = Q$$

$\quad 0 \qquad\qquad$ θ=90 $\quad$ θ=0 $\quad$ θ=90

$$\varepsilon (E)(4\pi r L) = \rho_L L$$

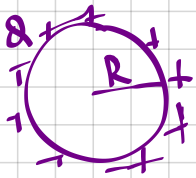
$$\boxed{E = \frac{\rho_L}{2\pi\varepsilon r} = \frac{Q}{2\pi\varepsilon r L}}$$

$Q = \int \rho_L \, dl$
$= \rho_L L$

$\rho \oint \bar{D} \cdot \bar{dS}$

$\int_V \nabla \cdot \bar{D}$

→ E due to conducting sphere



C1 : $r > R$

$$\oint \bar{D}\cdot d\bar{S} = Q_{en}.$$

$$\varepsilon \oint \bar{E}\cdot d\bar{S} = Q$$

$$\varepsilon E\cdot(4\pi r^2) = Q$$

$$E = \frac{Q}{4\pi\varepsilon r^2}$$

C2: $r = R$

$$\oint \bar{D}\cdot d\bar{S} = Q$$

$$\varepsilon\cdot E\cdot(4\pi R^2) = Q$$

$$E = \frac{Q}{4\pi\varepsilon R^2}$$

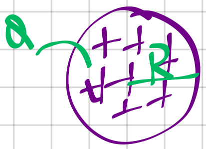
C3: $r < R$

$$\oint \bar{D}\cdot d\bar{S} = Q_{enc}$$

$$\varepsilon \oint \bar{E}\cdot d\bar{S} = 0$$

$$E = 0$$

→ E due to non conducting sphere



C1 : $r > R$

$$\oint \bar{D}\cdot d\bar{S} = Q_{enc}$$

$$\varepsilon\cdot E\cdot(4\pi r^2) = Q$$

$$E = \frac{Q}{4\pi\varepsilon r^2}$$

C2: $r = R$

$$\oint \bar{D}\cdot d\bar{S} = Q_{enc}$$

$$\varepsilon\cdot E\cdot(4\pi R^2) = Q$$

$$E = \frac{Q}{4\pi\varepsilon R^2}$$

C3: $r < R$

$$\oint \bar{D}\cdot d\bar{S} = Q_{enc}$$

$$\varepsilon\cdot E(4\pi r^2) = Q'$$

$$\left\{\begin{array}{l} \frac{4}{3}\pi R^3 \text{ --- } Q \\ \frac{4}{3}\pi r^3 \text{ --- } Q' \\ Q' = \frac{r^3 Q}{R^3} \end{array}\right\}$$

$$\varepsilon\cdot E\cdot(4\pi r^2) = \frac{r^3 Q}{R^3}$$

$$\boxed{E = \frac{Q r}{4\pi\varepsilon R^3} = \frac{\rho_v r}{3\varepsilon}}$$

→ Electric potential diff

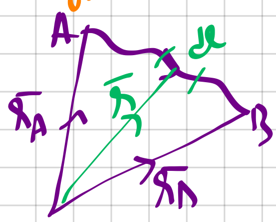
$$dW = -\bar{F}\cdot d\bar{l}$$ work done against field

$$dW = -q\bar{E}\cdot d\bar{l}$$

$$\int dW = -q\int_l \bar{E}\cdot d\bar{l}$$

$$\frac{W}{q} = -\int_l \bar{E}\cdot d\bar{l}$$

$$\boxed{V_{AB} = -\int_l \bar{E}\cdot d\bar{l} = V_B - V_A}$$



$V = \frac{W}{q}$

$W = FS$
$= qE\cdot S$

→ $V_{AB} = -\int \frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2}\hat{a_r}\cdot dr\,\hat{a_r}$

$$= -\int_{r_A}^{r_B} \frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2}\,dr$$

$$= \frac{Q}{4\pi\varepsilon_0}\left(\frac{1}{r}\right)\Big|_{r_A}^{r_B}$$

$$= \frac{Q}{4\pi\varepsilon_0}\left(\frac{1}{r_B} - \frac{1}{r_A}\right)$$

$$= \frac{kQ}{r_B} - \frac{kQ}{r_A}$$

$V = \frac{kQ}{R}$ $\quad E = \frac{kQ}{R^2}$

$$V_{AB} = V_B - V_A$$

$\bar{E} = \frac{kQ\bar{R}}{R^3}$

→ moving pt charge from $\infty \to r$

$$V_{AB} = kQ\left(\frac{1}{r} - \cancel{\frac{1}{\infty}}^{\,0}\right)$$

$$\boxed{V_{AB} = \frac{Q}{4\pi\varepsilon r} = \frac{kQ}{r}}$$

→ Relation b/w E & V

$$\boxed{\bar{E} = -\nabla V}$$

→ $V_{AB} + V_{BA} = 0$

$$-\int_A^B \bar{E}\cdot d\bar{l} - \int_B^A \bar{E}\cdot d\bar{l} = 0$$

$$\boxed{\oint_l \bar{E}\cdot d\bar{l} = 0}$$ Max wells 2nd eqn (integral form)

→ Wkt, $\oint_l \bar{A}\cdot d\bar{l} = \int_S (\nabla\times\bar{A})\cdot d\bar{S}$

(from stokes theorem)

$$\oint_l \bar{E}\cdot d\bar{l} = \int_S (\nabla\times\bar{E})\cdot d\bar{S} = 0$$

$$\boxed{\nabla\times\bar{E}=0}$$

Maxwells 2nd eqn (diff./pt form)

→ V due to point charge Q

$$V = E\cdot r = \frac{kQ}{r^2}\cdot r = \frac{kQ}{r}$$

$$\boxed{V_{AB}(\text{charge}) = kQ\left(\frac{1}{r_A}-\frac{1}{r_B}\right)}$$

+ C_1 pt charge

V_{AB}(line) ⇒

$$V = -\int \bar{E}\cdot dl$$

→ $V_B - V_A$

$$= -\int_{\rho_B}^{\rho_A} \frac{\rho_L}{2\pi\epsilon\rho}\,d\rho$$

$$= \frac{-\rho_L}{2\pi\epsilon}\ln\rho\Big|_{\rho_B}^{\rho_A}$$

$$\boxed{V_{AB} = \frac{-\rho_L}{2\pi\epsilon}\ln\frac{\rho_A}{\rho_B}}$$

+ C_2 line charge

→ Poissons & laplace eqn

$$\nabla\cdot\bar{D} = \rho_v$$

$$\nabla\cdot(\epsilon\bar{E}) = \rho_v$$

$$\nabla\cdot(\epsilon\cdot -\nabla\bar{V}) = \rho_v$$

$$\nabla\cdot\nabla\bar{V} = -\frac{\rho_v}{\epsilon}$$

$$\boxed{\nabla^2 V = \frac{-\rho_v}{\epsilon}}$$

→ Energy

$$W = \frac{1}{2}QV$$

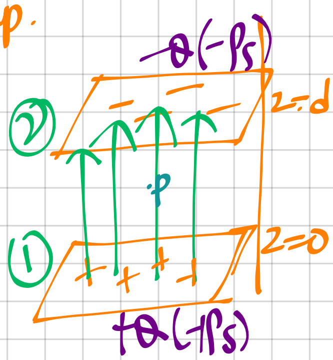
$$W_E = K\sum_{i<j}\frac{Q_iQ_j}{R_{ij}}$$

$$W = \frac{1}{2}\epsilon_0\int_V E^2\,dV$$

→ Capacitance

$$Q = CV$$

(1) parallel plate cap.



$$V_{21} = -\int_2^1 \bar{E}\cdot d\bar{l}$$

$$\bar{E}_p = \frac{\rho_s}{2\epsilon_0}\hat{a}_z + \frac{-\rho_s}{2\epsilon_0}(-\hat{a}_z)$$

$$E_p = \frac{\rho_s}{\epsilon_0}\hat{a}_z = \frac{Q}{A\epsilon_0}\hat{a}_z$$

$q = CV$

$C = \frac{q}{V}$

→ $V_{21} = -\int_2^1 \frac{Q}{A\epsilon_0}\hat{a}_z\cdot dz\,\hat{a}_z$

$$= -\frac{Q}{A\epsilon_0}\,z\Big|_{z=d}^{z=0}$$

$$V_{21} = \frac{Qd}{A\epsilon_0}$$

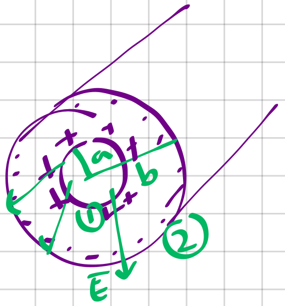
$$\boxed{C = \frac{Q}{V} = \frac{\epsilon_0 A}{d}}$$

With dielectric,

$$\boxed{C_{||} = \frac{\epsilon_0\epsilon_r A}{d}}$$

→ $\boxed{W_E = \frac{1}{2}CV^2 = \frac{1}{2}QV = \frac{1}{2}\frac{Q^2}{C}}$

(2) Coaxial cap.



$$V_{21} = -\int_2^1 \bar{E}\cdot \bar{dl}$$

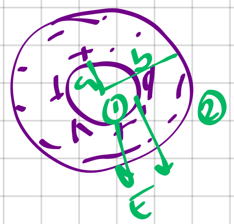
$$\left\{ E = \frac{Q}{2\pi\varepsilon r L}\,\hat{a}_r \right\}$$

$$V_{21} = -\int_{r=b}^{a} \frac{Q}{2\pi\varepsilon r L}\,\hat{a}_r \cdot dr\,\hat{a}_r$$

$$= -\frac{Q}{2\pi\varepsilon_0 L}\,\ln(r)\Big|_b^a$$

$$\boxed{V = \frac{Q}{2\pi\varepsilon_0 L}\ln\left(\frac{b}{a}\right) \qquad C = \frac{2\pi\varepsilon_0 L}{\ln\left(\frac{b}{a}\right)}}$$

(3) Spherical cap.



$$V_{21} = -\int_2^1 \bar{E}\cdot\bar{dl}$$

$$= -\int_b^a \frac{Q}{4\pi\varepsilon_0 r^2}\,\hat{a}_r \cdot dr\,\hat{a}_r$$

$$= -\frac{Q}{4\pi\varepsilon_0}\,\frac{-1}{r}\Big|_b^a$$

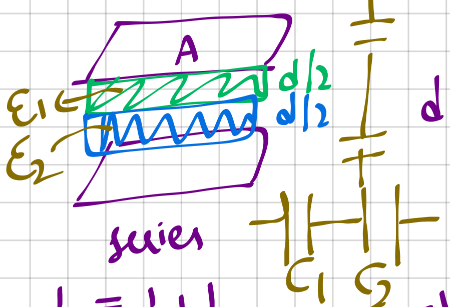
$$\boxed{V_{21} = \frac{Q}{4\pi\varepsilon_0}\left(\frac{1}{a}-\frac{1}{b}\right) \qquad C = \frac{4\pi\varepsilon_0}{\left(\frac{1}{a}-\frac{1}{b}\right)}} \quad \left(\frac{KQ}{a}\right)$$

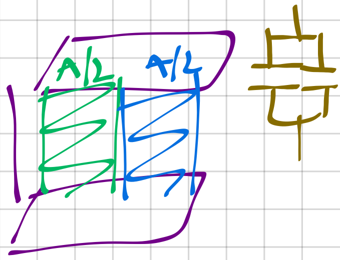
→ if outer at ∞ (isolated sphere)

$$\boxed{C = 4\pi\varepsilon_0 a}$$

→ Series & ||el

(1) ||el plate cap





series

$$\frac{1}{C'} = \frac{1}{C_1} + \frac{1}{C_2}$$

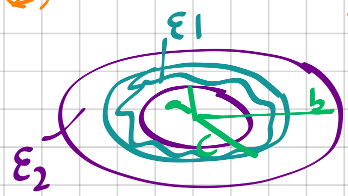
$$C_1 = \frac{\varepsilon_1 A}{d/2},\quad C_2 = \frac{\varepsilon_2 A}{d/2}$$

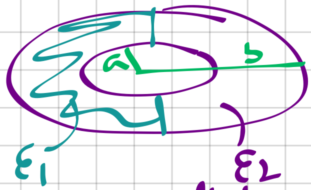
parallel

$$C' = C_1 + C_2$$

$$C_1 = \frac{\varepsilon_1 A/2}{d},\quad C_2 = \frac{\varepsilon_2 A/2}{d}$$

(2) coaxial cap





series

$$\frac{1}{C'} = \frac{1}{C_1} + \frac{1}{C_2}$$

$$C_1 = \frac{2\pi\varepsilon_1 L}{\ln\left(\frac{c}{a}\right)},\quad C_2 = \frac{2\pi\varepsilon_2 L}{\ln\left(\frac{b}{c}\right)}$$

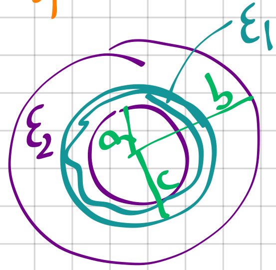
parallel

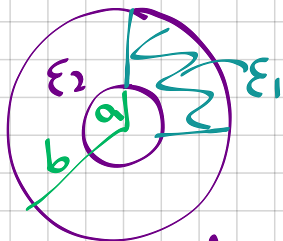
$$C' = C_1 + C_2$$

$$C_1 = \frac{1}{2}\,\frac{2\pi\varepsilon_1 L}{\ln(b/a)}$$

$$C_2 = \frac{1}{2}\,\frac{2\pi\varepsilon_2 L}{\ln(b/a)}$$

(3) spherical





Series

$$\frac{1}{C'} = \frac{1}{C_1} + \frac{1}{C_2}$$

$$C_1 = \frac{4\pi\varepsilon_1}{\frac{1}{a}-\frac{1}{c}},\quad C_2 = \frac{4\pi\varepsilon_2}{\frac{1}{c}-\frac{1}{b}}$$

parallel

$$C' = C_1 + C_2$$

$$C_1 = \frac{1}{4}\,\frac{4\pi\varepsilon_1}{\frac{1}{a}-\frac{1}{b}}$$

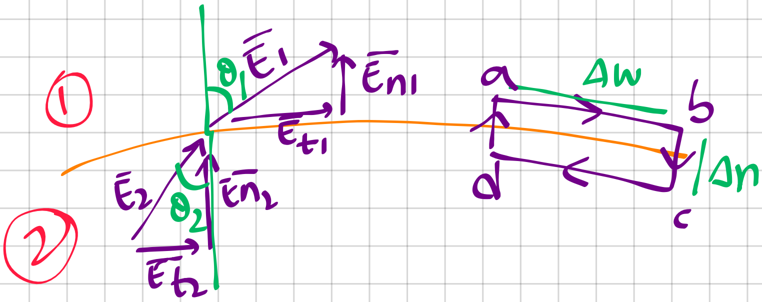
$$C_2 = \frac{1}{4}\,\frac{4\pi\varepsilon_2}{\frac{1}{a}-\frac{1}{b}}$$

→ Boundary conditions [BC]

→ Maxwells eqn:

1) $\oint_L \bar{E}\cdot\bar{dl} = 0$ 2

2) $\oint_S \bar{D}\cdot\bar{ds} = Q_{enc}$ 1



$\oint \bar{E}\cdot\bar{dl} = 0$

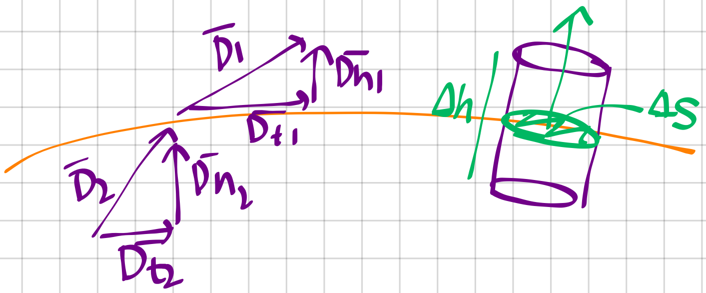
abrda

$\int_a^b + \int_b^c + \int_c^d + \int_d^a = 0$

$(E_{t1}\Delta w) + \left(-E_{n1}\frac{\Delta n}{2} - E_{n_2}\frac{\Delta n}{2}\right)$

$+ \left(-E_{t_2}\Delta w\right) + \left(E_{n1}\frac{\Delta n}{2} + E_{n_2}\frac{\Delta n}{2}\right) = 0$

$\boxed{E_{t_1} = E_{t_2}}$ 1st BC



$\oint \bar{D}\cdot\bar{ds} = Q_{enc.}$

$D_{t1}\,\Delta S\cos 90^\circ + D_{n1}\,\Delta S\cos 0^\circ$

$+ D_{t_2}\Delta S\cos 90^\circ + D_{n_2}\Delta S\cos 180^\circ = Q$

$(D_{n_1} - D_{n_2})\,\Delta S = \rho_s\,\Delta s$

$\boxed{D_{n1} - D_{n_2} = \rho_s}$

→ if no charge

$\boxed{D_{n1} = D_{n2}}$

$D_{n1} = D_{n_2}$

→ $\varepsilon_0\varepsilon_{r_1}E_{n1} = \varepsilon_0\varepsilon_{r_2}E_{n_2}$

$\boxed{\frac{E_{n1}}{E_{n_2}} = \frac{\varepsilon_{r_2}}{\varepsilon_{r_1}}}$

$\varepsilon_r E$

$D = \varepsilon_0\varepsilon_r E$

$E_{t1} = E_{t_2}$

→ $\frac{D_{t1}}{\varepsilon_0\varepsilon_{r1}} = \frac{D_{t_2}}{\varepsilon_0\varepsilon_{r_2}}$

$\boxed{\frac{D_{t1}}{D_{t_2}} = \frac{\varepsilon_{r1}}{\varepsilon_{r2}}}$

$E_{t_2} = E_2\sin\theta_2$ $\quad E_{t1} = E_1\sin\theta_1$

$E_{n_2} = E_2\cos\theta_2$ $\quad E_{n1} = E_1\cos\theta_1$

$E_{t1} = E_{t_2}$

$E_1\sin\theta_1 = E_2\sin\theta_2$

$\varepsilon_{r_1}E_1\cos\theta_1 = \varepsilon_{r_2}E_2\cos\theta_2$

$\boxed{\frac{\tan\theta_1}{\varepsilon_{r_1}} = \frac{\tan\theta_2}{\varepsilon_{r_2}}}$

$\tan\theta_1 = \frac{|E_{t1}|}{|E_{n1}|}$

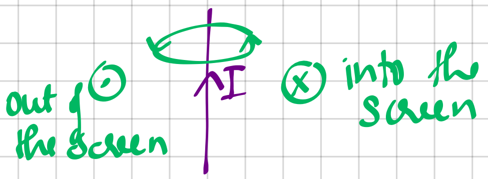
Magnetostatics

→ Biot - Savarts law

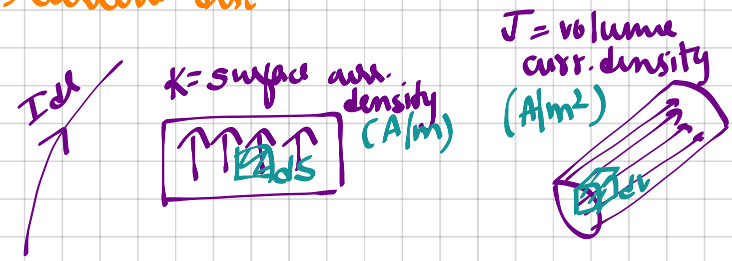
$$\boxed{\bar H = \frac{1}{4\pi}\,\frac{I\,\bar{dl}\times\bar R}{R^3}\quad \frac{A}{m}}$$

Dir^n of mag field intensity

↳ right hand thumb rule



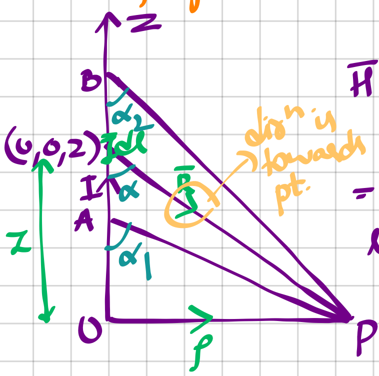
→ Current dist.



$$I\bar{dl} \equiv \bar K ds \equiv \bar J dv$$

$A \cdot m \qquad \frac{A}{m}\cdot m^2 \qquad \frac{A}{m^2}\cdot m^3$

→ Mag field intensity due to straight filamentous conductor



$$\bar H = \int_l \frac{I\,\bar{dl}\times\bar R}{4\pi R^3}$$

$$= \int_l \frac{I\left(dz\,\hat a_z \times [-z\hat a_z + \rho\hat a_\rho]\right)}{4\pi R^3}$$

$$\left\{\begin{array}{l}\bar{dl} = dz\,\hat a_z \\ \bar R = z(-\hat a_z) + \rho\hat a_\rho\end{array}\right\}$$

$$\{R^2 = z^2 + \rho^2\}$$

$$\left\{\begin{array}{l}\tan\alpha = \frac{\rho}{z} \\ z = \rho\cot\alpha \\ dz = -\rho\,\mathrm{cosec}^2\alpha\end{array}\right\}$$

$$\bar H = \int_l \frac{I\rho\,dz\,\hat a_\phi}{4\pi(z^2+\rho^2)^{3/2}}$$

$$= \int_\alpha \frac{-I\rho^2\,\mathrm{cosec}^2\alpha\;\hat a_\phi}{4\pi(\rho^2\cot^2\alpha+\rho^2)^{3/2}}$$

$$= \frac{-I\rho^2}{4\pi}\int_\alpha \frac{\mathrm{cosec}^2\alpha\,d\alpha\,\hat a_\phi}{\rho^3(1+\cot^2\alpha)^{3/2}}$$

$$= \frac{-I\rho^2}{4\pi\rho^3}\int_\alpha \frac{\mathrm{cosec}^2\alpha\,d\alpha\,\hat a_\phi}{\mathrm{cosec}^3\alpha}$$

$$\bar H = \frac{-I}{4\pi\rho}\int_{\alpha_1}^{\alpha_2} \sin\alpha\,d\alpha\;\hat a_\phi$$

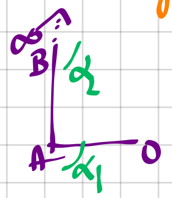
$$= \frac{-I}{4\pi\rho}\cdot -\cos\alpha\Big|_{\alpha_1}^{\alpha_2}\;\hat a_\phi$$

$$\boxed{\bar H = \frac{I}{4\pi\rho}(\cos\alpha_2 - \cos\alpha_1)\,\hat a_\phi}$$

ρ: ⊥er dist. from conductor to point of interest.

$\hat a_\phi = \hat a_l \times \hat a_\rho$

→ Semi infinite long

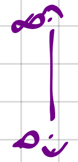


B (0 0 ∞) $\alpha_2 = 0°$

A (0 0 0) $\alpha_1 = 90°$

$$\boxed{\bar H = \frac{I}{4\pi\rho}\,\hat a_\phi}$$

→ Infinite long

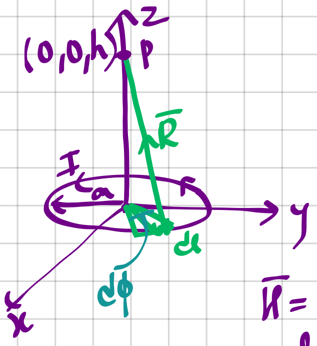


B (0 0 ∞) $\alpha_2 = 0°$

A (0 0 -∞) $\alpha_1 = 180°$

$$\boxed{\bar H = \frac{I}{2\pi\rho}\,\hat a_\phi}$$

→ $\bar H$ due to circular cond.



$$\bar H = \int_l \frac{I\,\bar{dl}\times\bar R}{4\pi R^3}$$

$$\left\{\begin{array}{l}\bar{dl} = a\,d\phi\,\hat a_\phi \\ \bar R = a(-\hat a_\rho) + h\hat a_z\end{array}\right\}$$

(Cancels out)

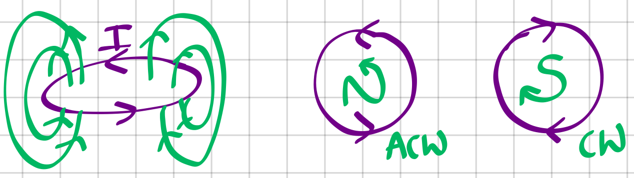
$$\bar H = \int_l \frac{I\left(a\,d\phi\,\hat a_\phi \times [-a\hat a_\rho + \cancel{h\hat a_z}]\right)}{4\pi R^3}$$

$$= \int_\phi \frac{I\,a^2\,d\phi\,\hat a_z}{4\pi(a^2+h^2)^{3/2}}$$

$$= \frac{I\,a^2\,\hat a_z}{4\pi(a^2+h^2)^{3/2}}\int_0^{2\pi} d\phi$$

$$\boxed{\bar H = \frac{I a^2\,\hat a_z}{2(a^2+h^2)^{3/2}}}$$

always the for Il in ACW dirn



→ Ampere's law

$$\oint_l \bar{H}\cdot\bar{dl} = I_{enc}$$

$$I_{enc} = \int_s \bar{J_c}\cdot\bar{ds}$$

$$\begin{cases} J = \frac{I}{A} \\ I = JA \end{cases}$$

→ from stokes, $\oint_l \bar{A}\cdot\bar{dl} = \int_s (\nabla\times\bar{A})\cdot\bar{ds}$

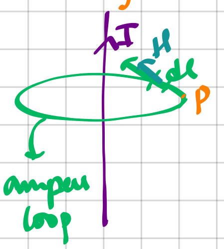
$$\Rightarrow \boxed{\oint_l \bar{H}\cdot\bar{ds} = \int_s \bar{J}\cdot\bar{ds}}$$ Maxwell's 3rd eqn (integral form)

$$\int_s (\nabla\times\bar{H})\,ds = \int_s \bar{J_c}\cdot\bar{ds}$$

$$\boxed{\bar{J_c} = \nabla\times\bar{H}}$$ Maxwell's 3rd eqn (diff. form)

$\bar{J_c} \to \bar{J_c}+\bar{J_D}$ (for time varying field.

→ Mag. field due to infinite long cond.



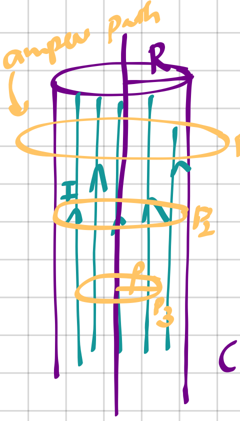
$$\oint \bar{H}\cdot\bar{dl} = I_{enc}$$

$$\int H\cdot dl\cdot\cos 0 = I$$

$$H(2\pi\rho) = I$$

$$\boxed{H = \frac{I}{2\pi\rho}}$$

→ $\bar{H}$ due to cond. of radius 'R':



C1: $\rho > R$

$$\oint_l \bar{H}\cdot\bar{dl} = I_{enc}$$

$$H(2\pi\rho) = I$$

$$\boxed{H = \frac{I}{2\pi\rho}}$$

C2: $\rho = R$

$$\oint \bar{H}\cdot\bar{dl} = I_{enc}$$

$$H(2\pi R) = I$$

$$\boxed{H = \frac{I}{2\pi R}}$$

C3: $\rho < R$

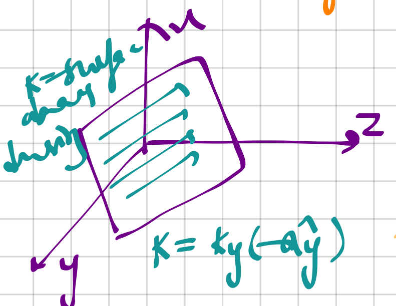
$$\oint \bar{H}\cdot\bar{dl} = I_{enc.}$$

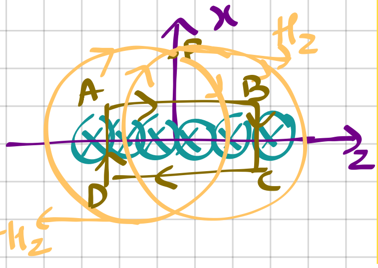
$$H(2\pi\rho) = I'$$

$$\boxed{H = \frac{I\rho}{2\pi R^2}}$$

$$\begin{cases} \pi R^2 \text{ — } I \\ \pi\rho^2 \text{ — } I' \\ I' = \frac{\rho^2}{R^2} I \end{cases}$$

→ $\bar{H}$ due to infinite curr. sheet





$$\oint \bar{H}\cdot\bar{dl} = I_{enc.}$$

$$\left(\int_A^B + \int_B^C + \int_C^D + \int_D^A\right)\bar{H}\cdot\bar{dl} = I_{enc}$$

$$\int_A^B H_2\cdot dl\cos 0 + \int_B^C H_2\cdot dl\cdot\cancel{\cos 90}^{\,0} + \int H_2\cdot dl\cos 0 + \int_D^A H_2\cdot dl\cdot\cancel{\cos 90}^{\,0} = I_{enc}$$

$$2H_2 L = k_y L$$

$$H_2 = \frac{k_y}{2}$$

$$\bar{H} = \begin{cases} k_y/2 & x > 0 \\ -k_y/2 & x < 0 \end{cases}$$

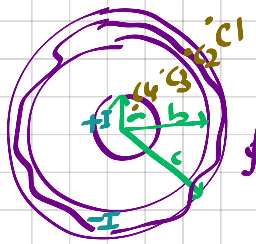
$$\Rightarrow \boxed{\bar{H} = \frac{1}{2}(\bar{k}\times\hat{a_n})}$$

unit normal vector from sheet to pt of interest

$x > 0$, $\bar{H} = \frac{1}{2}k_y\hat{a_z}$

$x < 0$, $\bar{H} = -\frac{1}{2}k_y\hat{a_z}$

→ $\bar{H}$ due to coaxial cond.



C1: $\rho > c$

$$\oint \bar{H}\cdot\bar{dl} = I_{enc}$$

$$\oint \bar{H}\cdot\bar{dl} = I - I$$

$$\boxed{H = 0}$$

C2: $b < \rho < c$

$$\oint \bar{H}\cdot\bar{dl} = I_{enc}$$

$$\begin{cases} \pi(c^2-b^2) \text{ — } -I \\ \pi(\rho^2-b^2) \text{ — } I' \\ I' = \frac{\rho^2-b^2}{c^2-b^2}(-I) \end{cases}$$

$$\int H\cdot dl\cos 0 = I + I'$$

$$H(2\pi\rho) = I\left(1-\frac{\rho^2-b^2}{c^2-b^2}\right)$$

$$\boxed{H = \frac{I}{2\pi\rho}\left(\frac{c^2-\rho^2}{c^2-b^2}\right)}$$

C3: $a < \rho < b$

$$\oint \bar{H}\cdot\bar{dl} = I_{enc.}$$

$$H(2\pi\rho) = I$$

$$\boxed{H = \frac{I}{2\pi\rho}}$$

C4: $\rho < a$

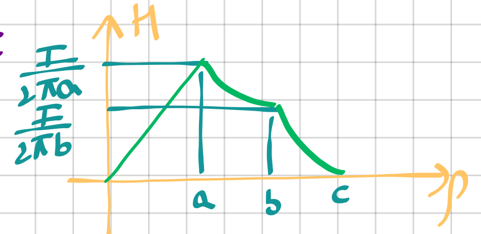
$$\oint \bar{H}\cdot\bar{dl} = I_{enc}$$

$$H(2\pi\rho) = I'$$

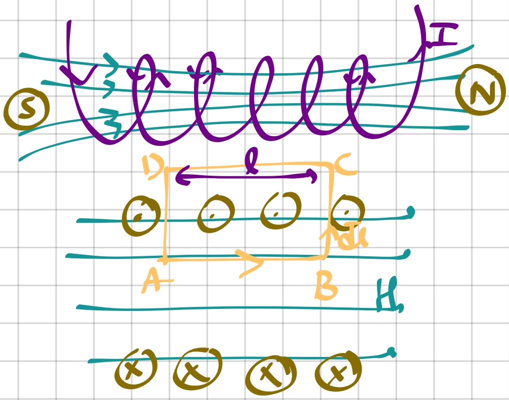
$$H(2\pi\rho) = \frac{\rho^2}{a^2} I$$

$$\boxed{H = \frac{I\rho}{2\pi a^2}}$$

$$\begin{cases} \pi a^2 \text{ — } I \\ \pi\rho^2 \text{ — } I' \\ I' = \frac{\rho^2}{a^2} I \end{cases}$$



→ $\bar{H}$ due to solenoid



cut from top to bottom in middle

$$\oint \bar{H}\cdot\bar{dl} = I_{enc}$$

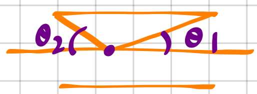
$$\int_A^B H\,dl\cos 0 + \int_B^C H\,dl\,\cancel{\cos 90}^{\,0} + \int_C^D \cancel{H\,dl\cos 180}^{\,0}$$

$$+\int_D^A H\,dl\,\cancel{\cos 90}^{\,0} = NI$$

$$H\cdot l = NI$$

$$\boxed{H = \frac{NI}{l}}$$

(#)



$$\boxed{H = \frac{nI}{2}(\cos\theta_1 + \cos\theta_2)}$$

→ $\bar{H}$ due to toroid

$$\oint_l \bar{H}\cdot\bar{dl} = I_{enc}$$

$$\oint H\cdot dl\cos 0 = NI$$

$$H(2\pi\rho) = NI$$

$$\boxed{H = \frac{NI}{2\pi\rho}}$$

→ Mag. flux density

$$\boxed{\bar{B} = \mu\bar{H}} \quad \left(\frac{Wb}{m^2} \text{ or Tesla}\right)$$

$\mu_0\mu_r$

$$\boxed{\psi = BA\cos\theta}$$

→ $$\boxed{\oint_S \bar{B}\cdot\bar{ds} = 0}$$ Maxwells 4th eqn (integral form)

From divergence theorem.

$$\oint_S \bar{B}\cdot\bar{ds} = \int_V (\nabla\cdot\bar{B})\,dV$$

$$\boxed{\nabla\cdot\bar{B} = 0}$$ Maxwells 4th eqn (diff. form)

TABLE

int. form	diff. form	Remarks
$\oint_S \bar{D}\cdot\bar{ds} = \int_V \rho_v\,dv = Q_{enc}$	$\nabla\cdot\bar{D} = \rho_v$	gauss law
$\oint_l \bar{E}\cdot\bar{dl} = 0$	$\nabla\times\bar{E} = 0$	W_{tot} in closed loop is 0.
$\oint_l E\cdot dl = \int_S \frac{\partial B}{\partial t}\,\bar{ds}$	$\nabla\times\bar{E} = -\frac{\partial B}{\partial t}$	faradays law
$\oint_l \bar{H}\cdot\bar{dl} = \int_S \bar{J}_c\cdot\bar{ds} = I_{enc}$ $\bar{J}_c + \bar{J}_D$	$\nabla\times\bar{H} = \bar{J}_c$ $\nabla\times\bar{H} = \bar{J}_c + \bar{J}_D$	amperes law
$\oint_S \bar{B}\cdot\bar{ds} = 0$	$\nabla\cdot\bar{B} = 0$	mag monopole doesnt exist.

→ Mag. scalar & vector potential.

→ vector identity :—

$\nabla C = 0$ $\quad \nabla\cdot(\nabla\times\bar{V}) = 0$

$CG = 0$ $\quad \nabla\times(\nabla V) = 0$

Maxwells :—

$\nabla\times\bar{H} = \bar{J}$

$\nabla\cdot\bar{B} = 0$

$E = -\nabla V$

$$\boxed{\bar{H} = -\nabla V_m} \boxed{\text{if } J = 0}$$

mag scalar pot.

→ $\bar{B} = \nabla\times\bar{A}$

mag. vector pot.

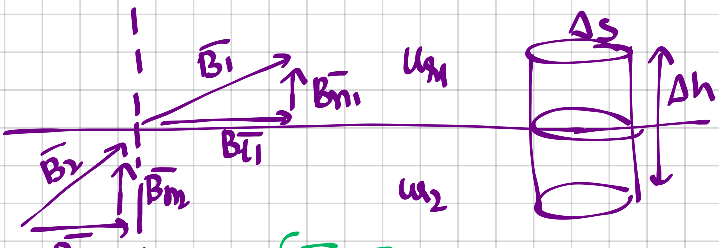
$$V = \int \frac{1}{4\pi\varepsilon_0}\frac{dQ}{R} \;;\quad \bar{A} = \int \frac{\mu_0}{4\pi}\frac{I\bar{dl}}{R} \quad \text{line curr.}$$

electrostatic

$$\bar{A} = \int \frac{\mu_0}{4\pi}\frac{\bar{K}\cdot ds}{R} \quad \text{surf. curr.}$$

$$\bar{A} = \int \frac{\mu_0}{4\pi}\frac{\bar{J}\cdot ds}{R} \quad \text{vol. curr.}$$

→ Mag boundary cond.



$$\oint_S \bar{B}\cdot d\bar{s} = 0$$

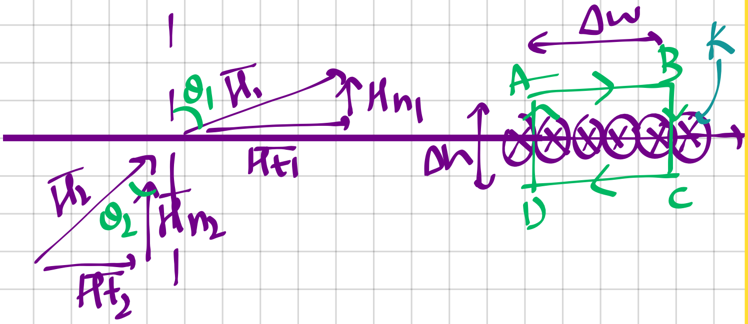
$$B_{n1}\,ds - B_{n2}\,ds = 0$$

$$\mu H = B$$

$$\boxed{B_{n1} = B_{n2}}$$

$$\mu_0\mu_{r1} H_{n1} = \mu_0\mu_{r2} H_{n2}$$

$$\boxed{\frac{H_{n1}}{H_{n2}} = \frac{\mu_{r2}}{\mu_{r1}}}$$



$$H_{t1}\Delta w - \cancel{H_{n1}\frac{\Delta h}{2}} - \cancel{H_{n2}\frac{\Delta h}{2}} - H_{t2}\Delta w$$

$$+\cancel{H_{n1}\frac{\Delta h}{2}} + \cancel{H_{n2}\frac{\Delta h}{2}} = I_{enc}$$

$$(H_{t1} - H_{t2})\Delta w = K\,\Delta w$$

$$\boxed{H_{t1} - H_{t2} = K}$$

$$\boxed{H_{t1} = H_{t2} \qquad K = 0}$$

$$\frac{B_{t1}}{\mu_{r1}} = \frac{B_{t2}}{\mu_{r2}}$$

$$\boxed{\frac{B_{t1}}{B_{t2}} = \frac{\mu_1}{\mu_2}}$$

→ $H_{t2} = H_2 \sin\theta_2$ $\quad$ $H_{t1} = H_1 \sin\theta_1$

$H_{n2} = H_2 \cos\theta_2$ $\quad$ $H_{n1} = H_1 \cos\theta_1$

$$\tan\theta_2 = \frac{H_{t2}}{H_{n2}} \qquad \boxed{\tan\theta_1 = \frac{H_{t1}}{H_{n1}}}$$

$$\boxed{\frac{\tan\theta_1}{\tan\theta_2} = \frac{\mu_1}{\mu_2}}$$

→ $M_1 = \chi_1 H_1$

$M_1 = (\mu_1 - 1) H_1$

→ poissons eqn

→ $\bar{B} = \nabla \times \bar{A}$

$\nabla \times \bar{H} = \bar{J}$ $\qquad$ $\{ B = \mu_0 H$

$\nabla \times \bar{B} = \mu_0 \bar{J}$

$\nabla \times (\nabla \times A) = \mu_0 \bar{J}$

$$\left[\nabla \times \nabla \times A = \nabla(\cancel{\nabla\cdot\bar{A}}) - \nabla^2 \bar{A}\right]$$

0

$$\Rightarrow \boxed{\nabla^2 \bar{A} = -\mu_0 \bar{J}}$$ Vector poissons eqn.

$$\nabla^2 V_m = 0$$

laplacian eqn

→ Derivation of Biot-savart and ampere circutal law:

Biot savart law:

$$\nabla \times (\nabla \times A) = \nabla.(\nabla \cdot A) - \nabla^2 A$$

$$B = \nabla \times \oint \frac{\mu_0 dl'}{4\pi R} = \frac{\mu_0 I}{4\pi}\oint \nabla \times \left(\frac{dl'}{R}\right) \quad F = dl', \; f = 1/R$$

$$= \frac{\mu_0 I}{4\pi}\oint\left[\frac{1}{R}\cancel{\nabla \times dl'} + (\nabla 1/R) \times dl'\right]$$

$\nabla \to x, y, z$

$dl' \to x', y', z'$

$$1/R = [(x-x')^2 + (y-y')^2 + (z-z')^2]^{-1/2}$$

$$\nabla\left(\frac{1}{R}\right) = \frac{(x-x')a_x + (y-y')a_y + (z-z')a_z}{|(x-x')^2 + (y-y')^2 + (z-z')^2|^{3/2}} = \frac{-a_R}{R^2}$$

$$B = \frac{\mu_0 I}{4\pi}\oint dl' \times \frac{a_R}{R^2}$$

Ampere circuital:

$$\oint_l \bar{H}\cdot d\bar{l} = \int_S (\nabla\times H)\cdot ds$$

$$= \frac{1}{\mu_0}\int_S \nabla\times(\nabla\times A)\cdot ds \quad = -\nabla^2 A$$

$$= \frac{1}{\mu_0}\cdot \mu_0\int J\cdot ds$$

$$\oint \bar{H}\cdot d\bar{l} = \int \bar{J}\cdot ds = I_{enc}.$$

unit 5 is ⟶

→ Magnetic force

(1) force on charge particle

$$\bar{F} = \bar{F}_E + \bar{F}_M$$

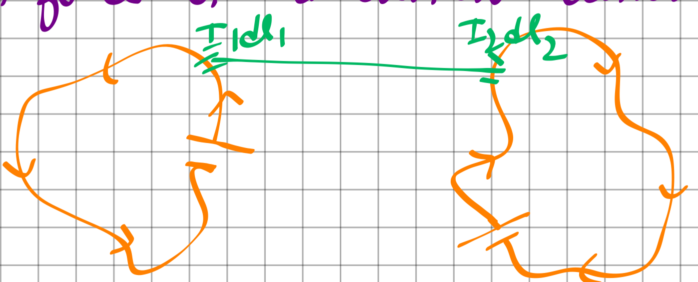
$$\boxed{\bar{F} = q\bar{E} + q(\bar{v}\times\bar{B})}$$ lorentz force eqn

stationary/moving — moving w/ velocity v.

(2) force on current element

$$\boxed{\bar{F} = I d\bar{l}\times\bar{B}}$$

(3) force b/w 2 current elements



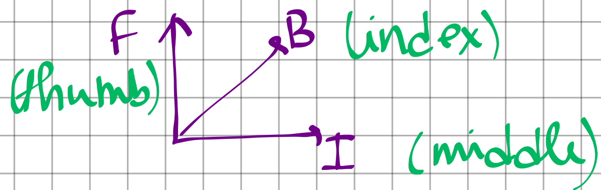
$$d(dF_1) = I_1 dl_1 \times d\bar{B}_2$$

$$\left\{ dB_2 = \frac{\mu_0}{4\pi}\int \frac{I_2 d\bar{l}_2 \times \hat{a}_{R_1}}{R_{21}^2} \right\}$$

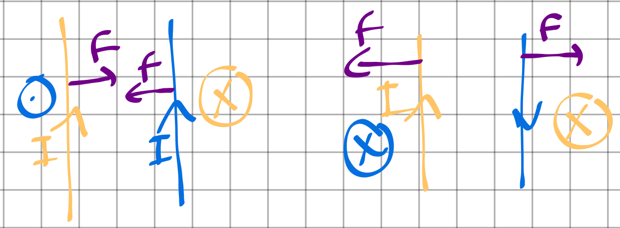
$$\boxed{F_1 = \frac{\mu_0}{4\pi} I_1 I_2 \oint_{l_1}\oint_{l_2} \frac{d\bar{l}_1 \times d\bar{l}_2}{R_{21}^2}\,\hat{a}_{R_{21}}}$$

→ dirn of force

flemmings left hand rule



→ eg.



→ Faradays law:

$$\boxed{V_{emf} = -\frac{\partial N\psi}{\partial t}}$$

mag flux

emf

lenz law

→ Lenz law:

dirn of flow of curr. is such that → induced mag. field produced by induced curr. will OPPOSE the change in original mag field

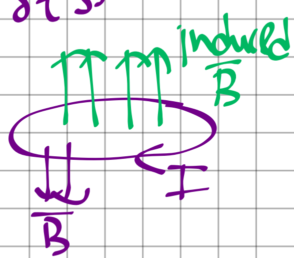
(1) loop is stationary (transformer emf)

$$V = \oint \bar{E}\cdot d\bar{l} = -N\frac{\partial}{\partial t}\int_S \bar{B}\cdot d\bar{s}$$

stokes, N=1

$$\int_S (\nabla\times\bar{E})\cdot ds = -\frac{\partial}{\partial t}\int_S \bar{B}\cdot d\bar{s}$$

$$\boxed{\nabla\times\bar{E} = -\frac{\partial\bar{B}}{\partial t}}$$



(2) loop is moving in static field (motional emf)

$$F_M = q(\bar{v} \times \bar{B})$$

$$\bar{E}_M = \frac{\bar{F}_M}{q} = \bar{v} \times \bar{B}$$

$$V_{emf} = \oint_l \bar{E} \cdot \bar{dl}$$

$$\boxed{V_{emf} = \oint_l (\bar{v} \times \bar{B}) \cdot \bar{dl}}$$

→ if $v \perp B$,

$$\boxed{\begin{aligned} F_m &= I l B \\ V_{emf} &= V l B \end{aligned}}$$

→ $\boxed{\nabla \times \bar{E} = \nabla \times (\bar{v} \times \bar{B})}$

(3) loop is moving in time varying field. (both transformer & motional emf present.)

$$V_{emf} = \oint_l \bar{E} \cdot \bar{dl}$$

$$= -\frac{\partial}{\partial t} \int_s \bar{B} \cdot \bar{ds} + \oint_l (\bar{v} \times \bar{B}) \cdot dl$$

$$\boxed{\nabla \times \bar{E} = -\frac{\partial B}{\partial t} + \nabla \times (\bar{v} \times \bar{B})}$$

→ Displacement Current

$$\boxed{J_D = -\frac{\partial D}{\partial t}} \qquad D = \varepsilon E$$

$$E = \frac{V}{d}$$

$$\boxed{I_D = J_D \cdot A}$$

Integral form: $\oint \vec{D} \cdot \vec{ds} = \int \rho v \cdot dv = Q_{enc}$ | $\oint H \cdot dl = \int J \cdot ds = I_{enc}$

Point form: $\nabla \cdot D = \rho v$ | $\nabla \times H = J$

$\nabla \cdot K$ → Gradient → scalar

$\nabla \times \vec{A}$ → curl

$\nabla \cdot \vec{A}$ → divergence

→ vector

$\vec{E} \to V/m$

$\vec{D} \to C/m^2$

$\vec{H} \to A/m$

$\vec{B} \to Wb/m^2$

sources are not changing their magnitude

Maxwell's final Time constant field equations:

(1) $\vec{\nabla} \times \vec{H} = J$	$\oint H \cdot dl = \int_s J \cdot ds$
(2) $\vec{\nabla} \times \vec{E} = 0$	$\oint_c E \cdot dl = 0$
(3) $\vec{\nabla} \cdot \vec{D} = \rho v$	$\oint_s \vec{D} \cdot ds = \int_v \rho v \, dv$
(4) $\vec{\nabla} \cdot \vec{B} = 0$	$\oint B \cdot ds = 0$

Lorentz Force Equation

$$\mathbf{F} = Q(\mathbf{E} + \mathbf{u} \times \mathbf{B})$$

Continuity Equation

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho_v}{\partial t}$$

Constitutive Relations

$$\mathbf{D} = \varepsilon \mathbf{E} = \varepsilon_0 \mathbf{E} + \mathbf{P}$$
$$\mathbf{B} = \mu \mathbf{H} = \mu_0(\mathbf{H} + \mathbf{M})$$
$$\mathbf{J} = \sigma \mathbf{E} + \rho_v \mathbf{u}$$

Boundary Conditions

$$E_{1t} - E_{2t} = 0 \quad \text{or} \quad (\mathbf{E}_1 - \mathbf{E}_2) \times \mathbf{a}_n = 0$$
$$H_{1t} - H_{2t} = K \quad \text{or} \quad (\mathbf{H}_1 - \mathbf{H}_2) \times \mathbf{a}_n = \mathbf{K}$$
$$D_{1n} - D_{2n} = \rho_s \quad \text{or} \quad (\mathbf{D}_1 - \mathbf{D}_2) \cdot \mathbf{a}_n = \rho_s$$
$$B_{1n} - B_{2n} = 0 \quad \text{or} \quad (\mathbf{B}_1 - \mathbf{B}_2) \cdot \mathbf{a}_n = 0$$

Perfect Conductor

$$\mathbf{E} = 0, \quad \mathbf{H} = 0, \quad \mathbf{J} = 0$$
$$B_n = 0, \quad E_t = 0$$

Compatibility Equations

$$\nabla \cdot \mathbf{B} = -\rho_m = 0$$
$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} = \mathbf{J}_m$$

Constitutive Equations

$$\mathbf{B} = \mu \mathbf{H}$$
$$\mathbf{D} = \varepsilon \mathbf{E}$$

Equilibrium Equations

$$\nabla \cdot \mathbf{D} = \rho_v$$
$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$